

QUESTIONS ABOUT LIFESTYLE

Living With Your ICD*



Restoring the rhythms of life.

Q How will I look after I have an ICD?

A To most people you will look the same as before the surgery. Your partner or family will be aware of the implant. The ICD's pulse generator is smaller than a deck of cards. You may see a small bulge under your skin.

The leads were either threaded through a vein to your heart or, in some cases, placed on the outside surface of your heart. The leads are quite thin and usually not visible.

You may have one or two incisions. You will have a scar at each incision. The scars should be less visible as time goes by.

Q Are there suggestions about comfortable clothing?

A Your normal attire should be comfortable for you. Men can wear suspenders or a belt depending on the implant site. Women can wear a loose belt, or skirts and slacks with a loose waistband. For pectoral (upper chest) implants, comfortable brassieres should be worn.

Q Will I feel comfortable?

A If you have an abdominally implanted ICD you may feel a slight tingling when you lean on a counter or other hard surface. Leaning on a counter does not harm the ICD. The tingling will not hurt you. You may also feel the weight of the pulse generator in some activities. For example, hunters, bowlers, or golfers may notice the weight of the ICD implanted in the upper chest area. This, too, may become less noticeable as time goes by.

Q Will people notice when the ICD delivers treatment?

A It depends on the arrhythmia and the treatment delivered.

Plan ahead about what to do if your ICD delivers treatment. If your ICD sends a shock to your heart, find a quiet place to sit or lie down. Ask a family member or friend to stay with you until you feel better. Ask your doctor for exact instructions about whether you should call after a therapy is delivered. Also talk to your family and friends about what they can do to help you.

Q How long will it take to adjust to my ICD?

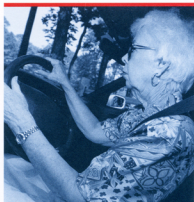
A That varies with each person. As you adjust to your ICD, you will begin to feel more comfortable. You have experienced many changes because of your tachyarrhythmia and ICD implant. It is normal for many emotions including relief, comfort, anxiety, fear, and anger to come and go over the course of many months. If you can talk about your concerns and feelings, it may be easier for you to adjust. Talk to someone you trust – your doctor, nurse, another ICD patient, or a good friend. Ask your doctor if there is an ICD patient support group near you. The goal is for you to lead a normal life as soon as possible.

Q How can I learn more about my ICD?

A Your doctor or nurse is best suited to answer questions about your ICD. In addition, if you would like more information about the Medtronic ICDs, contact Medtronic Patient Services at the phone number or address listed below.

QUESTIONS ABOUT DRIVING

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Q What are my responsibilities?

A Your responsibilities:

- Review your situation with your doctor noting his/her recommendations
- Investigate your state laws and insurance guidelines regarding driving with your heart condition and its treatment
- Follow your doctor's advice and comply with your individual state's guidelines

Q How can I learn more about my ICD?

A Your doctor or nurse is best suited to answer questions about your ICD. In addition, if you would like more information about Medtronic ICDs, contact Medtronic Patient Services at the phone number or address listed below.

Q Are patients with an ICD allowed to drive?

A The decision can vary by patient and state. Many ICD patients are able to resume driving privileges based on a positive health assessment and no likelihood of losing consciousness.

Q How is it determined whether it is safe to drive?

A The determination is made from the results of your electrophysiology (EP) tests, how you respond to the ICD and its therapies, your medications, and other variables. Remember, it isn't your ICD that affects driving safety, but rather your heart rhythm disorder (tachyarrhythmia) that may make you feel dizzy or cause you to faint. Resuming driving may not be safe for you and others. Your state's laws or your doctor may ask you to forgo driving for a certain length of time in order to assess your response to the ICD treatment of your tachyarrhythmia.

Q Are there state and insurance guidelines that will affect this decision?

A The laws and guidelines differ from state to state. It is important to research your individual state's guidelines and then talk them over with your physician. The laws and insurance guidelines in your state may require your doctor to inform these agencies or you of the risks to yourself and the public from your driving.

After discussion with your physician, it is best to follow your doctor's advice because it may not be safe for you to drive even if it is legal. Further, your insurance coverage could be an issue if you have been advised to give up driving.

Q How will I get by if I can't drive?

A Some options to consider:

- Investigate public transportation like buses, taxis, trains, subways
- Sometimes walking or biking are possible for short trips
- Sharing transportation with family, friends, neighbors, or work associates

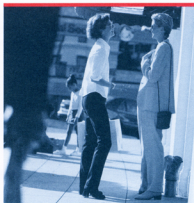
Give yourself some time to get used to your ICD. Most patients find the ICD can give you more freedom to return to normal daily activity.

Q Can my driving restrictions change?

A Maybe. Your doctor is concerned about your health and also the safety of other drivers. Regular contact with your physician will give you an opportunity to discuss your medical progress. You can also ask your doctor or nurse about driving safety guidelines or any state restrictions.

QUESTIONS ASKED BY THOSE WHO LIVE WITH AN ICD PATIENT

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Q Why was an ICD recommended?

A Your family member or friend has a heart rhythm disorder that causes the heart to beat too fast. An ICD can detect abnormal heart rates and deliver therapy to slow down the heart rate and return the heart to its normal rhythm.

Some patients use medication as well as an ICD. It is important to follow the physician's directions if medication is prescribed.

Q Will driving be an issue with an ICD?

A Some ICD patients have driving restrictions. When your family member's or friend's heart beats too fast or the ICD delivers a shock, they may feel dizzy or even pass out. If they pass out while driving, they may cause or be injured in an accident.

The doctor will talk to your family member or friend about safety and driving.

Q Will having an ICD create concerns regarding travel?

A Most patients feel more secure with an ICD and are able to travel freely. It is normal for ICD patients to worry about being far from emergency care. With the proper precautions, most ICD patients can travel safely.

After your family member or friend adjusts to the ICD, they will feel more comfortable about being away from home.

Q How can I help with the adjustment to an ICD?

A The goal is for the ICD patient to return to normal activities and lifestyle.

- Learn as much as possible about the ICD and the treatment it provides. For example, review all ICD take-home information (brochures, emergency care letters, etc.).
- Find out what to do if the ICD delivers shock therapy
- You may also wish to learn CPR (cardiopulmonary resuscitation)
- Ask if you should stay during routine follow-up visits
- Keep emergency phone numbers nearby

Additionally, talk to your family member or friend about their ICD. Listen to their fears and concerns. Encourage them to find the support they need with family, friends, or other ICD patients. Find out if there is a patient support group in your area.

Q What should I do when the ICD delivers a shock?

A Stay with the patient and make them as comfortable as possible. The ICD is doing what it is supposed to do. Talk to them. *If the patient does not respond or passes out, call your local emergency number or 911. If you know CPR and your family member's or friend's heart stops beating, begin CPR immediately. When their heart starts to beat again, stop CPR. In all cases, follow the physician's directions. If directed, call the physician's office for further instructions.*

Q How can I learn more about an ICD?

A Your family member's or friend's doctor or nurse is best suited to answer questions about the ICD. In addition, if you would like more information about Medtronic ICDs, contact Medtronic Patient Services at the phone number or address listed below.

Q What if I'm out of the country?

A Your doctor or nurse may have information about international heart centers. They may call the manufacturer for referrals as well. Otherwise, call a major medical institution when you reach your travel destination.

Referral: _____

Phone: _____

ICD Clinic: _____

Notes: _____

Q How can I learn more about my ICD?

A Your doctor or nurse is best suited to answer questions about your ICD. In addition, if you would like more information about Medtronic ICDs, contact Medtronic Patient Services at the phone number or address listed below.

Q Now that I have an ICD, will I be able to travel safely?

A Yes. Most ICD patients travel freely. Your ICD will deliver therapy if your heart beats too fast or too slow. This may help you feel more secure when you are away from home.

It is important to discuss travel plans with your doctor or nurse. They will tell you what precautions to take and give you instructions for care in case of an emergency.

Q Are there precautions to keep in mind?

A Yes.

- At the airport, show your ICD identification card and ask for a hand search. Airport screening devices may detect the ICD's metal case and set off the alarm. Your ICD will function normally, however. Any repetitive motions of the handheld scanning tool may interfere temporarily with ICD function.
- Walk normally through theft detection systems at entrances to stores, libraries, or museums. Do not stop in areas adjacent to detection equipment.

QUESTIONS ABOUT TRAVELING

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Q What can I do if I go on long trips?

A Some suggestions are:

- Carry your ICD identification card
- Wear your medical jewelry
- Ask your doctor to give you an emergency care sheet including the ICD settings. This sheet has important information that explains how to care for you in an emergency.
- Plan for emergencies (for example, if your device shocks you repeatedly or you don't feel well). Tell your traveling companions about your ICD. Tell them what to expect and what to do if your ICD delivers a high-energy shock or if you pass out.
- Ask your doctor if there is a heart center at your travel destination. Ask who to call if you have questions or have an emergency.
- If you are not near a heart center, call other emergency services to find out who can provide care for ICD patients

Q What if I'm out of town during my scheduled follow-up visit?

A Routine visits are very important. If you plan to be away from home for two or more months, ask your doctor to help you arrange for follow-up care at your travel destination.

PATIENT REGISTRATION AND IDENTIFICATION CARDS

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Q Does my ICD need to be registered and tracked?

A Yes, the Food and Drug Administration (FDA) Device Tracking Regulation says that certain implantable medical devices must be tracked from the manufacturer to the attending physician and patient. This enables manufacturers to notify physicians of any relevant information concerning your ICD. An ICD is one of approximately 20 devices covered by this regulation.

Q How is my ICD tracked?

A The process begins at the time of surgery when a registration form is completed and sent to the ICD manufacturer. At Medtronic, the registration information becomes part of a computer database that we continually update to ensure the information is accurate and current.

Q What information is on the registration form?

A It includes:

- The device's model and serial numbers
- Your name, address, and phone number
- Your social security number, if available
- Date your ICD was implanted
- The hospital where the implant was performed
- The implanting doctor's name, address, and phone number
- Your attending doctor's name, address, and phone number

Q Why do you need my social security number?

A It is a key piece of information and enables us to locate your address if for some reason we lose touch with you.

Q Is my registration information confidential?

A Yes. It is used for tracking your devices.

Q What happens if I move or change doctors?

A Please notify us as soon as possible. Your cooperation in keeping your information current will help us to better serve you and your doctor. Medtronic has a special patient toll-free telephone line for your convenience. It is 1 (800) 551-5544. If you prefer writing, our address is:

Q When do I receive my ID card?

A After the registration information is typed into our database, an identification card (ID card) and an emergency information card are mailed to you. The ID card lists your name and address, the model numbers of your ICD and leads, the serial number of the ICD, and your attending physician's name and telephone number. The card also has a picture of an implanted ICD to help you pass airport security checkpoints. The emergency card lists important information about treating an ICD patient in an emergency situation.

Q Should I carry my identification card all the time?

A Yes. Your ID card is designed to be carried with you so that this pertinent information is available if there is a situation that requires medical care. You may wish to invest in medical jewelry as well.

The six-language card (English, French, German, Spanish, Italian, and Japanese) below may help you when you are traveling overseas. Use it with your ID card to clear airport security.

If you do not have your card, your physician or nurse can call Medtronic or the Medical Records Department of the hospital where the implant took place for the information. Call Medtronic later for a replacement card.

Q How can I learn more about my ICD?

A Your doctor or nurse is best suited to answer your questions. In addition, if you would like more information about Medtronic ICDs, contact Medtronic Patient Services at the phone number or address listed below.

Q What is an ICD?

A An ICD is an implantable cardioverter defibrillator. It is used to treat heart rhythm disorders in which your heart beats too fast (ventricular tachyarrhythmias-VT). Your heart rhythm may also be unstable and irregular (ventricular fibrillation-VF). When the ICD detects VT or VF, it sends electric impulses to your heart. These impulses can return your heart to a normal rhythm. Your ICD can reduce your chances of cardiac arrest.

In addition, your Medtronic ICD is also similar to a pacemaker, in that it constantly monitors your heart and delivers support to your heart if it beats too slow.

Q How does my ICD work?

A Your ICD is placed inside your body during surgery. It is set up to send electric impulses as soon as your heart begins beating irregularly (VT or VF). The ICD will try several times, if necessary, to return your heartbeat to normal.

Medtronic ICDs deliver three types of treatment: pacing, cardioversion, and defibrillation.

■ **Pacing** – Your ICD may send pacing pulses (ATP) antitachycardia pacing when your heartbeat is just a little too fast. Several pacing pulses occur rapidly. Usually you won't feel them. You may feel lightheaded or dizzy because of your tachyarrhythmia. (Your device may not have ATP therapy if it is not effective with your heart rhythm disorder. Ask your doctor about your ICD.)

■ **Cardioversion** – Your ICD can send stronger electric impulses. These are called cardioversion shocks. Cardioversion shocks feel like a "thump on the chest." Again, you may feel lightheaded or dizzy because of your tachyarrhythmia.

■ **Defibrillation** – Your ICD sends a high-energy defibrillation shock if your heartbeat becomes unstable and irregular, or if previous therapies were not effective. If this happens, many patients pass out. Patients who have been awake during a defibrillation shock say it feels like a "kick in the chest." The shock is over quickly.

■ **Long-term support pacing therapy** – Sometimes after a shock is delivered, your heart rate may be too slow. If that happens, or if there is any other reason your heart may slow down, your ICD sends pacing pulses, like a pacemaker, that raise your heart rate to normal.

QUESTIONS ABOUT THERAPY

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Q Will the ICD cure my heart disease?

A No. Your ICD treats your heart rhythm disorder. It cannot prevent your heart from beating too fast. It can't cure or prevent the heart disease that may be the cause of your tachyarrhythmia.

Q How can I learn more about my ICD?

A Your doctor or nurse is best suited to answer questions about your ICD. In addition, if you would like more information about Medtronic ICDs, contact Medtronic Patient Services at the phone number or address listed below.

Q What should I do if my ICD delivers a shock?

A Ask your doctor or nurse to fill in the blanks below:

■ If my ICD shocks once and I feel fine _____

■ If my ICD shocks and I don't feel well _____

■ If my ICD shocks multiple times, my family should _____

QUESTIONS ABOUT ACTIVITIES AND INTIMACY

Living With Your ICD*



Q Now that I have an ICD, when will I be able to return to normal activities?

A The goal is for you to lead a normal life as soon as possible. Your ICD is implanted to treat your rhythm disorder and return it to a more normal pattern. The ICD has been compared to having a “paramedic” with you at all times. Hopefully, it will add a comfort level as you resume your normal lifestyle. As a general rule, use common sense as you start or resume physical activities. In addition, there are suggested precautions to follow during your recovery period.

Q What are the suggested precautions?

A *During the first month:*

- Avoid lifting more than 10-15 pounds
- Avoid excessive pushing, pulling, or twisting

During the first three months:

- Refrain from participating in heavy use of the arms in exercising or reaching (concerns upper chest implants)
- Avoid activities that could jeopardize your safety (i.e., climbing, swimming, boating alone, or camping in remote areas)
- Avoid driving until given an okay by your doctor

Ongoing:

- Avoid rough physical contact that includes jarring or falling
- Avoid playing football, baseball, soccer, and skiing if it includes rough physical contact
- Avoid hunting if a rifle butt is rested on the same side as an upper chest implant

If in doubt about an activity, always discuss with your doctor or nurse.

Q What activities can I resume?

A Generally, your ICD will not keep you from most activities and hobbies (i.e., bowling, golfing, playing tennis, gardening, fishing, etc.)

Q Will I be able to return to work?

A Most patients are able to return to work after heart surgery. You and your doctor will make this decision. The timing will depend on many things, including the type of work you do.

Q Will I be able to resume sexual activity?

A Yes. Intimacy is a normal part of life. Most patients resume sexual activity when their incisions are healed. There are many forms and degrees of intimacy including sexual intercourse, hugging, kissing, and cuddling. You and your partner should be able to enjoy all the benefits of intimacy.

Q Will my ICD deliver a shock during intimacy or daily activities?

A It is rare. Your ICD is programmed to allow your heart rate to normally rise without treatment shocks being delivered. Your device will only deliver a shock when your heart rate reaches the programmed upper rate (trigger).

For example, if your ICD is set at a trigger rate of 150 beats per minute (bpm), it will deliver a shock when your heart beats faster than 150 bpm.

Ask your doctor what your trigger rate is.

It is possible that intimacy and other activities could raise your heart rate so that the ICD delivers a shock.

Please notify your doctor or nurse if this occurs. There may be treatment options such as:

- Simply reprogramming the ICD's trigger rate
- Drug therapy to slow down the heart rate during exercise

Q Will my ICD hurt my partner if it shocks me during intimacy?

A No. Your partner may feel the shock or muscle spasm. This can be startling, but will not hurt your partner.

Q How can I learn more about my ICD?

A Your doctor or nurse is best suited to answer questions about your ICD. In addition, if you would like more information about Medtronic ICDs, contact Medtronic Patient Services at the phone number or address listed on the next page.